Amiodarone

Mechanism of action

- Amiodarone is an antidysrhythmic with a broad spectrum of activity.
- Amiodarone has predominantly class III activity. It prolongs the action potential duration, reduces automaticity and prolongs the refractory period of atrial, nodal and ventricular tissues.
- The electrophysiological effects result in a reduction in abnormal electrical activity (for example ectopy), a reduction in electrical conduction, a reduction in heart rate and a stabilisation of the SA and AV nodes.
- Amiodarone also causes a small increase in coronary blood flow (although this is not usually clinically significant) and a reduction in myocardial oxygen consumption by reducing inotropy (the force of cardiac contraction).

Authorised practice levels

- Paramedics: cardiac arrest.
- ICPs and CCPs: all indications.

Indications

- ✓ Cardiac arrest with VF or VT after three shocks.
- ✓ Adults with sustained VT in the absence of cardiac arrest.
- ✓ Adults with moderate cardiovascular compromise as a result of atrial fibrillation, where the ventricular rate is predominantly greater than 130/minute and metoprolol PO is contraindicated.
- ✓ Adults with moderate cardiovascular compromise as a result of atrial flutter, where the ventricular rate is predominantly greater than 130/minute.

Contraindications

- ✗ Known severe allergy.
- ✗ Known severe allergy to iodine.
- ✗ VT secondary to cyclic antidepressant poisoning. In this setting amiodarone administration can be associated with severe worsening of shock, without resolution of the rhythm.
- ✗ If torsades de pointes is present or has occurred.

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Cautions

- None if the patient is in cardiac arrest (with the exception of arrests involving tricyclic antidepressant poisoning or Torsades de Pointes, see contraindications).
- Poor perfusion or signs of low cardiac output. Amiodarone reduces inotropy and may cause a significant fall in cardiac output, particularly when administered rapidly.
- Hypotension. Amiodarone causes vasodilation and may worsen hypotension, particularly when administered rapidly.
- Atrial fibrillation associated with sepsis. Amiodarone may cause a significant fall in cardiac output.
- Known sick sinus syndrome without an internal pacemaker in place. Amiodarone slows the heart rate and severe bradycardia may occur following reversion of a tachydysrhythmia.
- Previous 2nd or 3rd degree heart block without an internal pacemaker in place. Amiodarone slows the heart rate and severe bradycardia may occur following reversion of a tachydysrhythmia.
- Pregnancy.

Use in pregnancy or when breastfeeding

- May cause harm during pregnancy. Do not administer amiodarone unless there is a strong clinical indication.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- Cardiac arrest:
  - 300 mg for an adult.
  - If in VF or VT after six shocks, a second dose of 150 mg may be administered.
  - See the paediatric drug dose tables for a child.
- Tachydysrhythmia in an adult:
  - 300 mg IV over 30 minutes.
  - A further 150 mg IV over 30 minutes may be administered if the tachydysrhythmia persists.

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Administration

- Cardiac arrest: administer IV undiluted as a bolus.
- Tachydysrhythmia:
  - Place 300 mg of amiodarone in 100 ml of 5% glucose and label. 1 drop/second via a standard IV administration set will deliver 100 ml over approximately 30 minutes. Slow the rate of infusion if hypotension occurs.
  - The administration set will need to be flushed with 0.9% sodium chloride to ensure that all of the amiodarone has been administered.
  - An IV infusion over 30 minutes is the preferred method of administration. However, it is acceptable to dilute 300 mg of amiodarone to a total volume of 20-30 ml using 5% glucose or 0.9% sodium chloride. Administer this IV over 30 minutes and slow the rate of infusion if hypotension occurs.

Common adverse effects

- Hypotension.
- Lightheadedness.
- Bradydysrhythmia.

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 1-4 hours after a single dose.
- Amiodarone is taken up into tissues and slowly released. This may result in a prolonged half-life, particularly when more than one dose has been administered. This is why many references quote a half-life of 10-60 days, but the clinical duration of effect is much shorter than this.

Usual preparation

- Ampoule containing 150 mg in 3 ml.

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Pharmacokinetics

- Amiodarone is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- May potentiate the action of cyclic antidepressants in cyclic poisoning.
- May cause bradycardia following reversion of dysrhythmia if the patient is taking a beta-blocker and/or a centrally acting calcium channel blocker (for example diltiazem).

Additional information

- If the indication is atrial fibrillation causing moderate cardiovascular compromise, the goal of treatment is to control the ventricular rate and not to revert the rhythm to sinus rhythm, although treatment with amiodarone may result in reversion of the rhythm to sinus rhythm. If a patient has been in atrial fibrillation for longer than a few days, there is a small risk that this may be associated with emboli leaving the left atrium. This is why amiodarone is reserved for patients with cardiovascular compromise that is clinically significant.
- If amiodarone is commenced, the full dose should be administered even if the rhythm reverts to sinus rhythm, unless severe hypotension or bradycardia occurs.
- Amiodarone is often described as relatively contraindicated in the presence of a prolonged QT interval, but this only applies to long-term administration.
- When diluting amiodarone with 0.9% sodium chloride the solution will go slightly cloudy. This does not adversely affect the amiodarone.
- Avoid shaking the ampoule because this will cause the solution to froth.

More Information

Version history

16 June 2026 1.1.0.1

Update: Indications, contraindications and dosage updated.